D.1 Guided Cantilever Method

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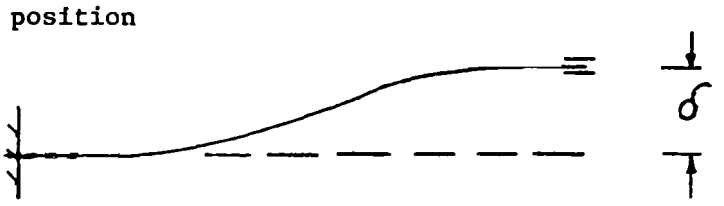
BASIC EQUATION

Definition of "Guided Cantilever"

A beam with one end fixed and the other end held parallel to its original position



Maximum moment at either end, M (in-lbf)

$$M = 6PL$$

Maximum deflection, δ (in)

$$\delta = \frac{144\,PL^3}{EI}$$

Maximum stress, S_b (psi)

$$S_b = \frac{Mc}{I} = \frac{MD}{2I}$$

Solving for stress in terms of deflection

$$S_b = .0208\,\frac{\delta ED}{L^2}$$

where

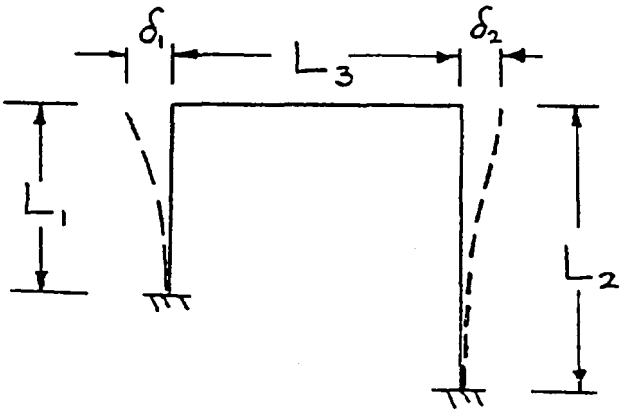
D	-	outside diameter of pipe, in
E	-	pipe material modulus of elasticity at installation temperature, psi. The ANSI B31.3 says "Bending and torsional stresses shall be computed using the as-installed modulus of elasticity ... ". The terms "cold modulus" can lead to confusion in cryogenic systems.
L	-	length of pipe being deflected, ft.

PROPORTIONED DEFLECTION

Many times the thermal expansion of one segment of a piping system is absorbed by several perpendicular segments. The following example shows the deflection of L_3 being absorbed by L_1 and L_2. The amount of deflection absorbed by each leg can be found.

Total deflection, Δ

$$\Delta = \delta_1 + \delta_2$$



$$\Delta = \frac{144\ PL_1^3}{EI} + \frac{144\ PL_2^3}{EI}$$

assuming constant pipe cross-section

$$\frac{\delta_1}{\Delta} = \frac{144PL_1^3\ /(EI)}{144P/(EI)\ (L_1^3 + L_2^3)} = \frac{L_1^3}{L_1^3 + L_2^3}$$

$$\delta_1 = \frac{L_1^3}{L_1^3 + L_2^3}\Delta \qquad \delta_2 = \frac{L_2^3}{L_1^3 + L_2^3}\Delta$$

If the deflection is shared by N legs, the deflection of the ith leg is given by

$$\delta_i = \frac{L_i^3\ \Delta}{\sum_{n=1}^{N} L_n^3}$$

COMBINING STRESSES FOR GUIDED CANTILEVER

In general, any one section or leg of pipe calculated for stress using the Guided Cantilever method will absorb deflection from the two directions perpendicular to this section of pipe. These two separate deflections will give two separate stress values. These two values of stress, in theory, should be combined; but in practice do NOT need to be combined.

Case for not combining

When using the Basic Method of guided cantilever stress calculation, the stresses should not need to be combined. Use only the largest of the two stress values to compare with the stress limit (15,000 for most cases). The following two points show why.

1. When combining bending stresses per B31.3 we use

$$S_b = \sqrt{S_{b1}^2 + S_{b2}^2}$$

The maximum amount by which the combined stress can exceed the largest component is 41%. This occurs when the two components are equal. In most cases there is a significant difference between components, and the combined stress will nearly equal the largest component.

(smallest component) / (largest component)	(combined stress) / (largest component)
1.0	1.41
0.9	1.35
0.8	1.28
0.7	1.22
0.6	1.17
0.5	1.12

2. When a stress limit of 15,000 is used, the maximum amount of actual stress could be 21,200 psi. This value is well below the code value for allowable stress range for most steels. If you are using a steel whose code value of allowable stress range is below 21,200, divide the allowable stress range by 1.41 and use the resulting value in place of 15,000.

If necessary to combine

When a more accurate value of stress is required, such as when using the more sophisticated refinements of the Guided Cantilever method, the stresses can be combined using

$$S_b = \sqrt{S_{b1}^2 + S_{b2}^2}$$

Combining Stresses for Guided Cantilver (continued)

S_b - combined bending stress

S_{b1} - first component of bending stress

S_{b2} - second component of bending stress

Limitations of Guided Cantilever Method

Constant moment of inertia

a) For each leg

b) For all legs sharing deflection
If the legs sharing a deflection do not have the same moment of inertia, the deflection proportioning equation must be altered.

Between Adjacent Restraints

The only exception is when any intermediate restraints do not restrict the movements being considered.

Assumes Excessive Restraints

A one way stop is analyzed as an anchor.

Limited to Relatively Simple Geometries
Stress due to bending of a leg in two perpendicular directions can be calculated with the resultant stress.

USE OF BASIC EQUATION

The stress in the ith leg due to a perpendicular deflection is

$$S = .0208 \frac{\delta_i ED}{L_i^2}$$

δ (in), E (psi), D (in), L (ft) For the basic method limit S_{bi} to 15,000 psi for steel unless a higher limit is acceptable to the PM Principal Engineer.

for carbon steel pipe installed at 70 deg F

$$S_{bi} = 5.8 \times 10^5 \frac{\delta_i D}{L_i^2}$$

If the minimum length of the ith leg is required

$$L_i = 0.144 \sqrt{\frac{\delta_i ED}{S_b}}$$

for carbon steel installed at 70 deg F with a 15,000 psi stress limit

$$L_i = 6.21 \sqrt{\delta_i D}$$

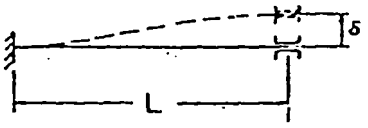
If the maximum deflection of the ith leg is required

$$\delta_i = 48.1 \frac{S_b L_i^2}{ED}$$

for carbon steel installed at 70 deg F with a 15,000 psi stress limit

$$\delta_i = .0259 \frac{L_i^2}{D}$$

Guided Cantilever Chart



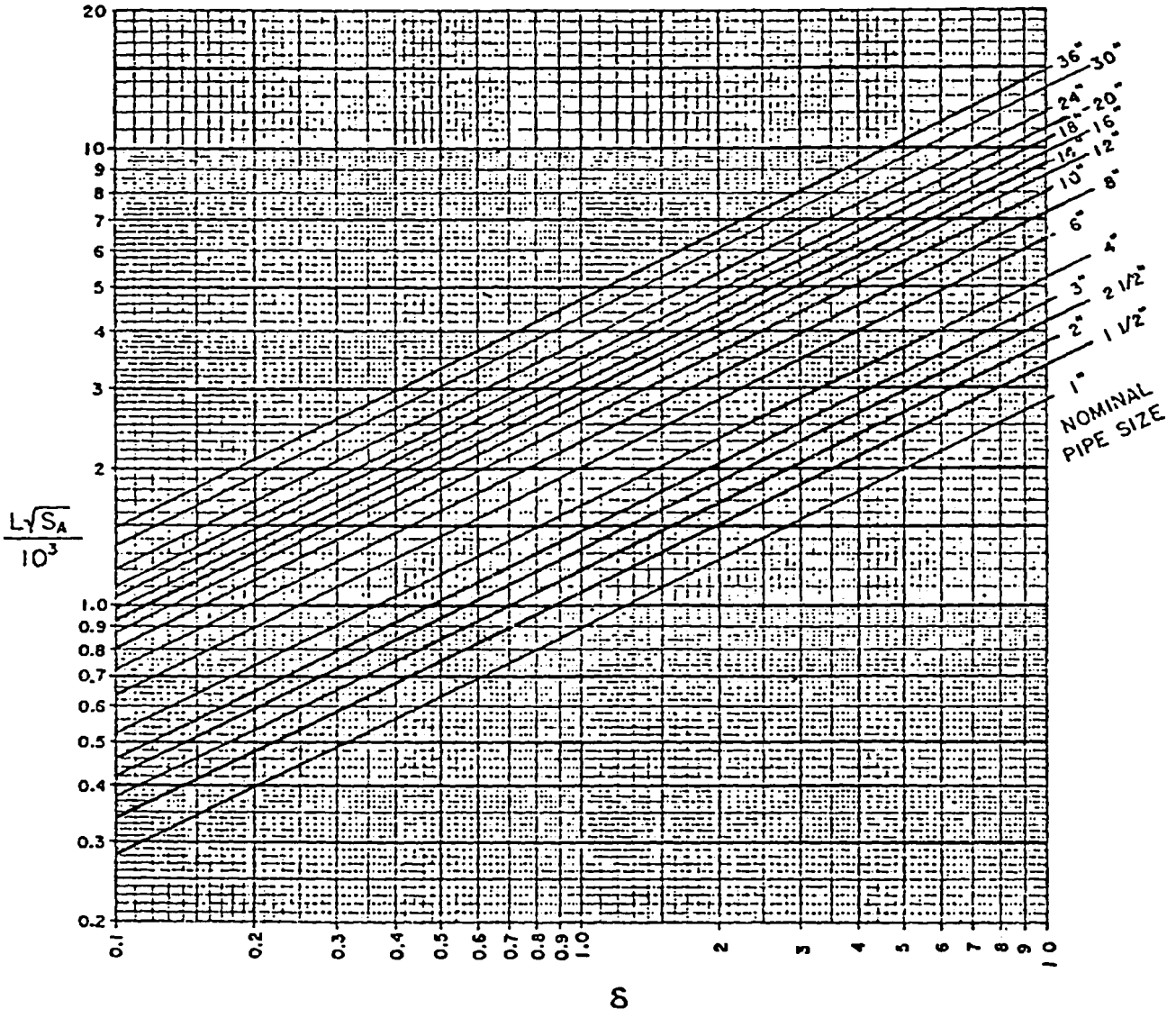
Assumed mode of deflection of guided cantilever.

L = Length of leg, ft.

δ = Lateral deflection, in.

Value of E used = 29×10^6 psi.

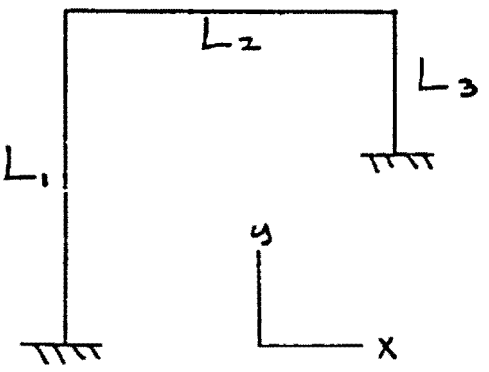
S_A = Code allowable stress range ($1.25S_c + 0.25S_h$), psi.



Instructions: Determine value of $L\sqrt{S_A}/10^3$. Enter with this value of ordinate scale and read over to line for proper nominal pipe size. Read down to abscissa scale. The value obtained will be the permissible lateral deflection for leg.

This chart neglects CF and S.I.F.

EXAMPLE PROBLEM USING THE BASIC METHOD



4" sch 40 pipe
carbon steel at 400°F

L_1 = 10 ft., L_2 = 10 ft., L_3 = 5 ft.

find e = .027 in/ft
D = 4.5 in

calculate $\Delta x = eL_2$ = .027 (10) = .27 in.
$\Delta y = e(L_1 - L_3)$ = .027 (10-5) = .135 in.

Notice that leg L_2 is only deflected by the differential expansion between L_1 and L_3.

To find the proportional deflection for each leg

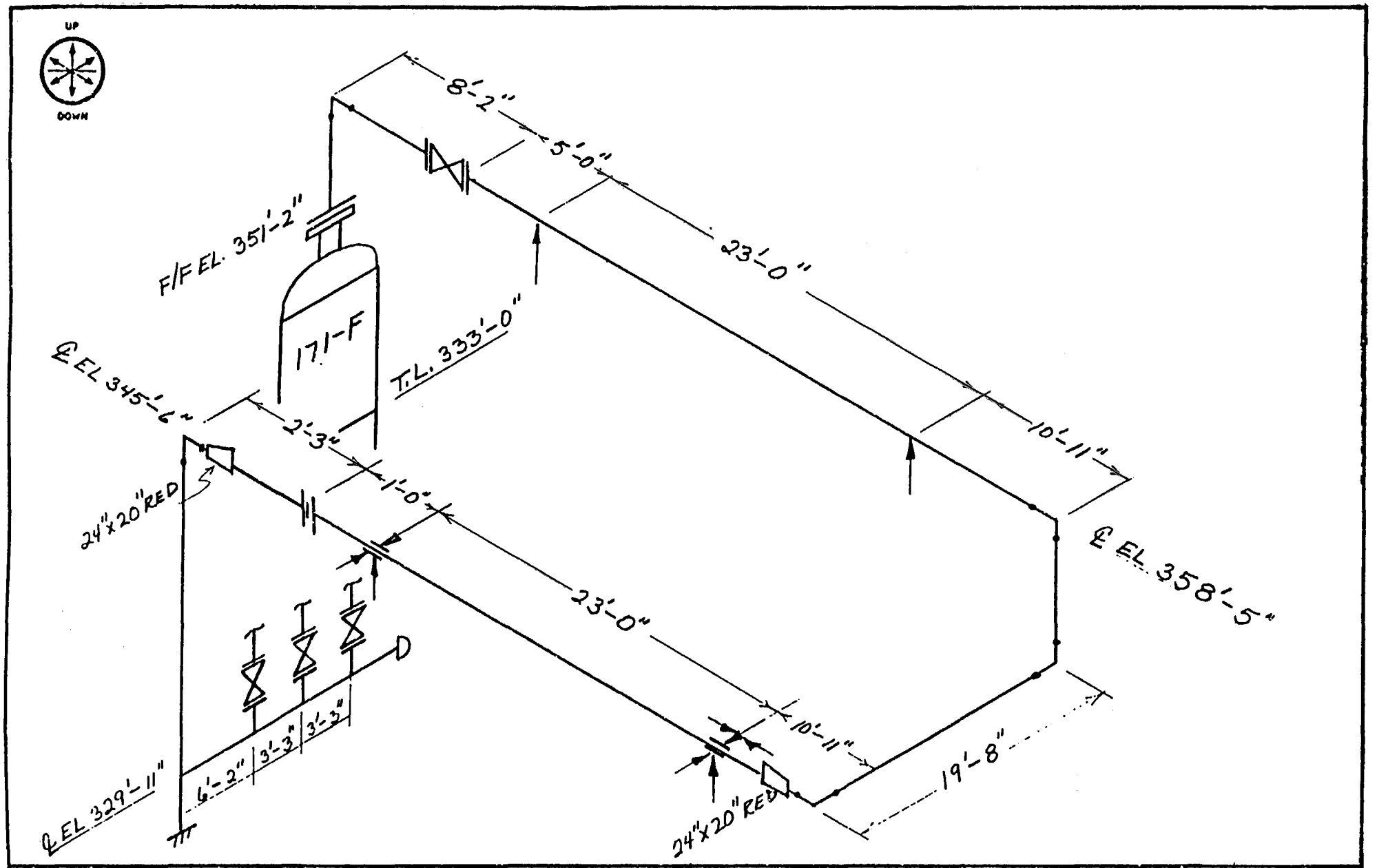
$$\delta = \Delta \left(L^3 / \Sigma L^3 \right)$$

To find the stress in each leg

$$S_b = 5.8 \times 10^5 \, \delta D / L^2$$

For the X direction Stress member	L	$L^3/\Sigma L^3$	δ	S
L_1	10	.89	.24	6264
L_3	5	.11	.03	3132
For the Y direction				
L_2	10	1.0	.135	3524

Notice that the wall thickness has no part in the calculations. In practice wall thickness has a very secondary effect on stress.



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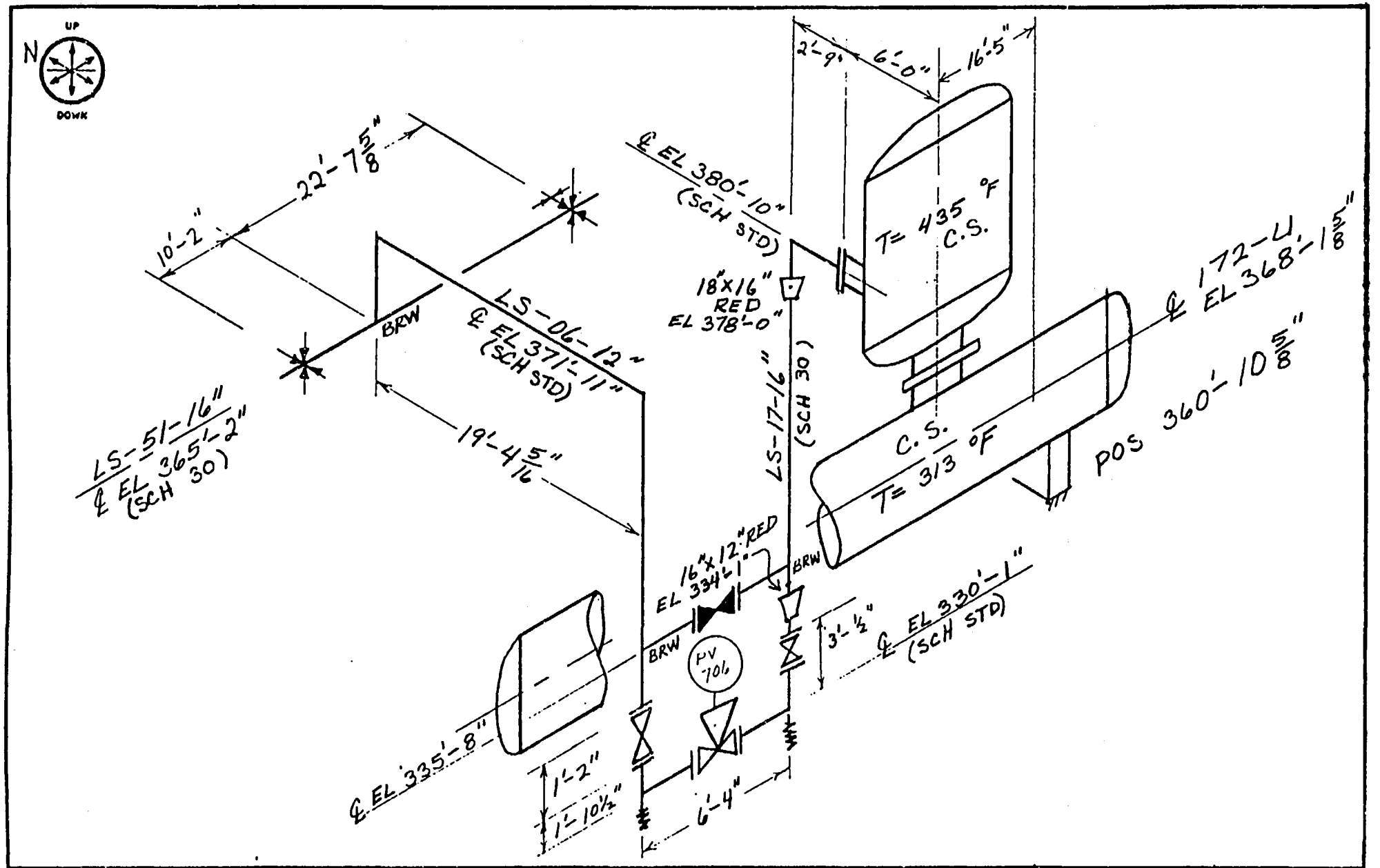
DR		450/20	FG-02-24"	SCH 20			A53-B	ISSUED FAD	Pullman Kellogg Division of Pullman Incorporated	
CH		450/20	FG-02-20"	SCH 20			A53-B	ISSUED CONST		
SUPT		250	171-F				STAINLESS	APP'D	M	
	FIELD SUPT	FLEX TEMP	LINE NO	CLASS	DESIGN TEMP	DESIGN PRESS	LINE MAT'L		CLASS / AREA	JOB NUMBER / DRAWING NO



REV. NO	1	CH BY		DATE	2	CH BY		DATE	3	CH BY		DATE	4	CH BY		DATE

DR		550	LS-06-12"				A53-B	ISSUED FAB	Pullman Kellogg Division of Pullman Incorporated
CH		550	LS-17-16"				"	ISSUED CONST	
SUPT		550	LS-51-16"				"	APP D	M
	FIELD SUPT	FLEX TEMP	LINE NO	CLASS	DESIGN TEMP	DESIGN PRESS	LINE MAT'L		CLASS AREA / JOB NUMBER / DRAWING NO

FRICTION

Friction is normally not considered in the thermal expansion stress calculations. The friction force on axial restraints or pipe guides due to long runs of pipe are normally calculated by hand. In most thermal stress calculations the effect due to friction is rather arbitrary and makes little difference to the accuracy of the calculations.

The exception to this philosophy is when pipes are connected to pieces of sensitive equipment. In these cases the effect due to friction can significantly change the results. In general, this would include only large compressors and turbines.

When friction must be considered for lines of 24" pipe size or smaller, the effect of friction is considered only to the first support on horizontal pipe away from the nozzle. When there are weight supports on pipe which is not horizontal between the nozzle and the first support on horizontal pipe the friction effects on these intermediate supports will also be considered. On lines with pipe sizes greater than 24" the effect of friction is considered at all weight support points and all other restraint points from the nozzle out to a point where these effects will no longer influence the nozzle loads.

The techniques for modelling weight friction are given in section V.F.5533. Computer modelling of friction load should not be done on guides and stops where hand calcualtion methods can adequately calculate these forces.